AIML Transaction & Fraud Analysis

Business Insight & Executive Report | Prepared by Analytics Team

Dashboard Overview

This business intelligence report synthesizes core monitoring metrics, transaction volumes, exposure balances, and anomaly signals from the updated Power BI transaction dashboard. The platform incorporates automated transaction logging with real-time rule engines flagging suspicious activity and categorizing structural fraud patterns.

TOTAL TRANSACTIONS	FRAUDULENT TXNS	FRAUD RATE
1M (~1.32M)	**2K (1,719)**	**0.13%**

TOTAL ACCOUNTS	TOTAL TXN VOLUME	AVG TXN AMOUNT
10K	**153.48bn**	**115.99K**

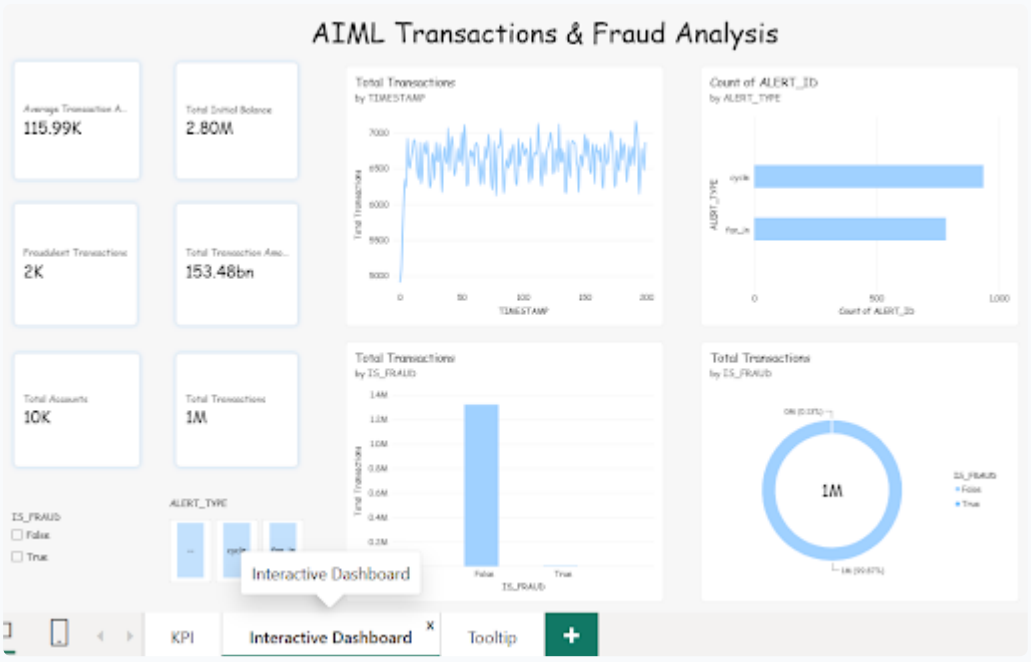


Figure 1. Primary interactive dashboard overview featuring global KPIs, time-series trends, alert distribution, and class split.

1. KPI and Fraud Exposure Analysis

A rigorous review of the visible card visuals reveals key foundational telemetry across account populations, individual transfer volumes, and total processed value.

KPI Metric	Dashboard Display	Analytical Interpretation
Total Transactions	1M (Formatted)	Exact raw count is ~1.32M transactions across the recorded period.
Fraudulent Transactions	2K (Formatted)	Total transactions flagged with `IS_FRAUD = True` (exact count: 1,719).
Total Accounts	10K	Unique account entities active within the monitored system.
Total Transaction Value	153.48bn	Cumulative currency volume transferred across all channels.
Average Transaction	115.99K	Mean transfer value per transaction event.
Total Initial Balance	2.80M	Combined starting opening balances across all 10K accounts.

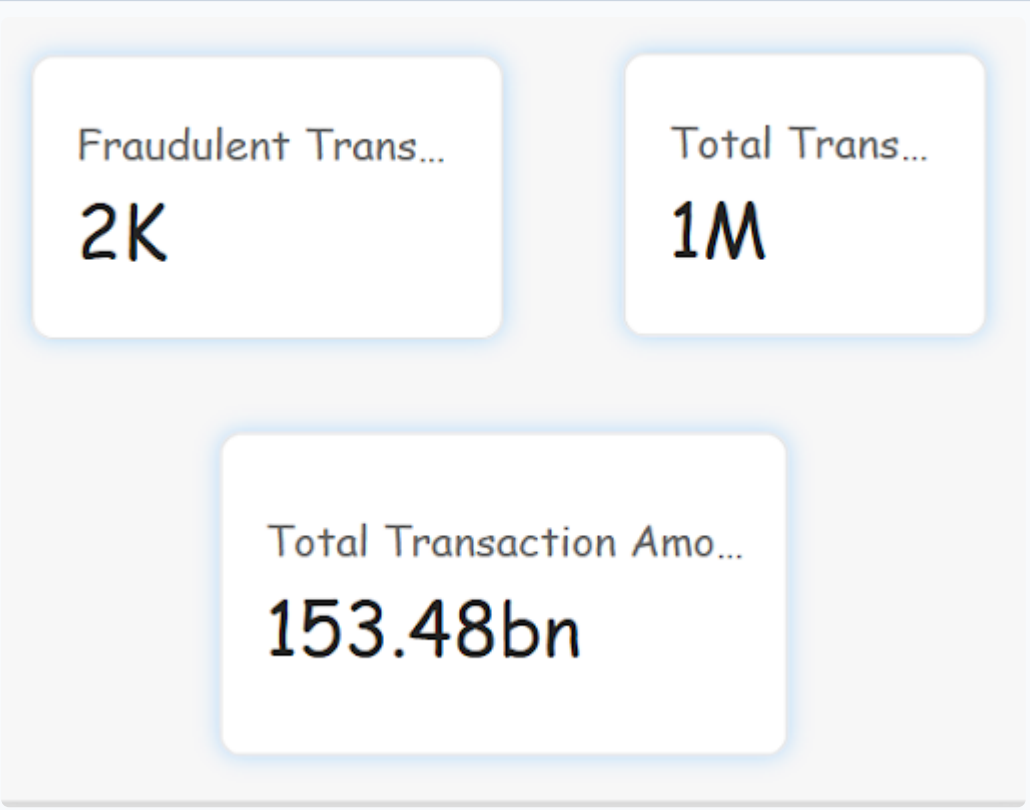


Figure 2. Focused view on primary metric cards showing formatted thresholds (Fraudulent Txns: 2K, Total Txns: 1M, Amount: 153.48bn).

DATA CONSIDERATIONS & INTEGRITY NOTES

- **Synthetic Ratio Anomaly:** Total transaction volume ($153.48bn) significantly outweighs total initial balances ($2.80M), indicating high transaction velocity or synthetic simulation data.
- **Financial Loss Unstated:** Dashboard metrics display fraud instance counts but do not isolate cumulative monetary fraud loss

2. Transaction Trend and Alert Typology

Evaluating temporal transaction distribution alongside rule-based alert triggering mechanisms provides critical context on system throughput stability and operational alert workloads.

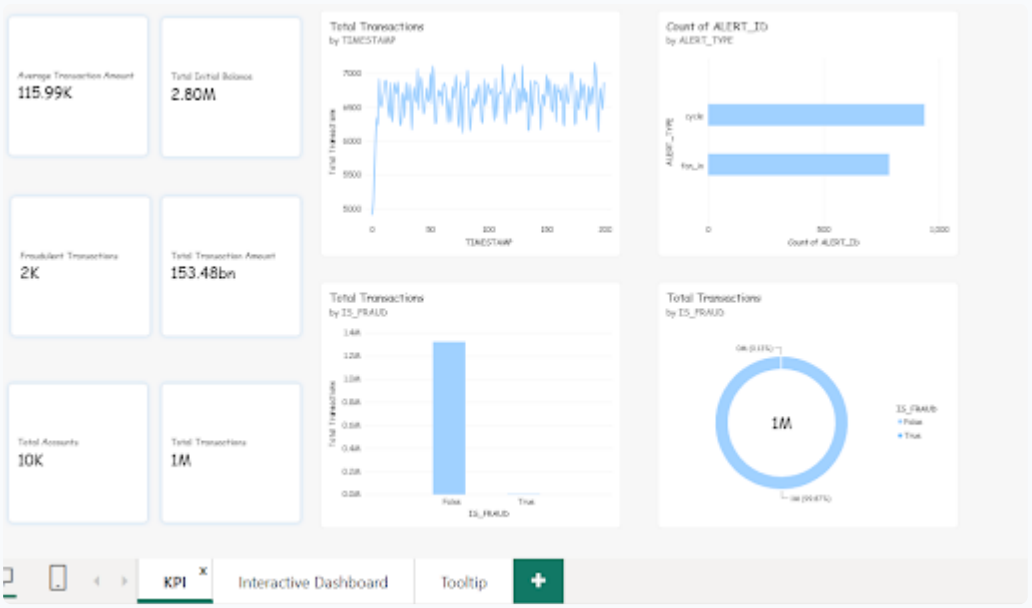


Figure 3. Full interactive dashboard view displaying time-series stability and alert count distribution.

Temporal Transaction Dynamics

The time-series plot (*Total Transactions by TIMESTAMP*) shows an initial ramp-up from ~5,000 to ~6,500 transfers at timestamp 0, reflecting a system warm-up phase. Past this initial stage, throughput fluctuates stably between 6,000 and 7,200 transactions per interval across 200 time steps, exhibiting regular variance without macro upward or downward drift.

Alert Breakdown & Typology

System alerts are partitioned into two principal topologies: **Cycle** (~950 occurrences) and **Fan_In** (~780 occurrences). Combined, these alerts total ~1,730 cases, closely tracking the total flagged fraud volume of 1,719. This strongly indicates that *Cycle* and *Fan_In* represent primary fraud detection typologies.

3. Fraud Profile and Class Imbalance

Analyzing class distribution is essential when evaluating machine learning detection models. Highly skewed datasets require specific evaluation metrics to avoid misinterpreting model efficacy.

Class Distribution Metrics

Out of approximately 1.32 million total transactions, **1,719** are confirmed fraudulent (0.13%), while **99.87%** represent legitimate transfers. On standard vertical bar charts, the fraud column is nearly invisible relative to the legitimate volume bar.

Implications for AI/ML Modeling

A naive classifier predicting all transfers as non-fraudulent would achieve a benchmark accuracy of **99.87%** while failing to catch any fraudulent activity. Consequently, accuracy is an unsuitable metric; validation must focus on Precision, Recall, PR-AUC, and F1-Score.

Slicer & Filtering Diagnostics

The dashboard includes interactive slicers for `IS_FRAUD` (True/False) and `ALERT_TYPE` (`--`, `cycle`, `fan_in`). The presence of the `--` designation suggests unclassified transactions or benign events generating no system alerts. Verification is recommended to rule out data pipeline logging gaps.

4. Business Insights & Recommendations

Based on empirical observations across transaction trends, alert concentrations, and balance ratios, the following strategic actions are recommended:

Observed Pattern	Strategic & Operational Recommendation
Severe Class Imbalance Fraud accounts for 0.13% (1,719 txns).	**Model Calibration:** Implement synthetic oversampling (SMOTE), cost-sensitive learning, or class-weighted loss functions in model training. Measure success via Precision-Recall AUC rather than ROC-AUC or Accuracy.
Alert Typology Convergence Cycle (~950) & Fan_In (~780) equal total fraud.	**Targeted Rule Engineering:** Optimize rule thresholds specifically targeting circular payment loops (*Cycle*) and rapid aggregation patterns (*Fan_In*).
Initial Ramp-up Volatility Volume jumps from 5K to 6.5K at Timestamp 0.	**Baseline Exclusions:** Exclude the initial initialization window (Timestamp 0–10) from baseline statistical calculations to avoid skewed alert triggers.
High Velocity to Balance Ratio $153.48bn volume vs $2.80M opening balance.	**Data Pipeline Validation:** Conduct schema validation to confirm currency scales and ensure simulation parameters align with operational real-world conditions.

5. Conclusion and Data Notes

Conclusion: The monitored Power BI dashboard provides an effective high-level overview of enterprise transaction volume (~1.32M) and risk signals. It confirms that fraud incidents remain tightly bound within specific detection patterns (*Cycle* and *Fan_In*). Further enhancements in feature reporting—specifically isolating net monetary loss—will improve strategic decision-making.

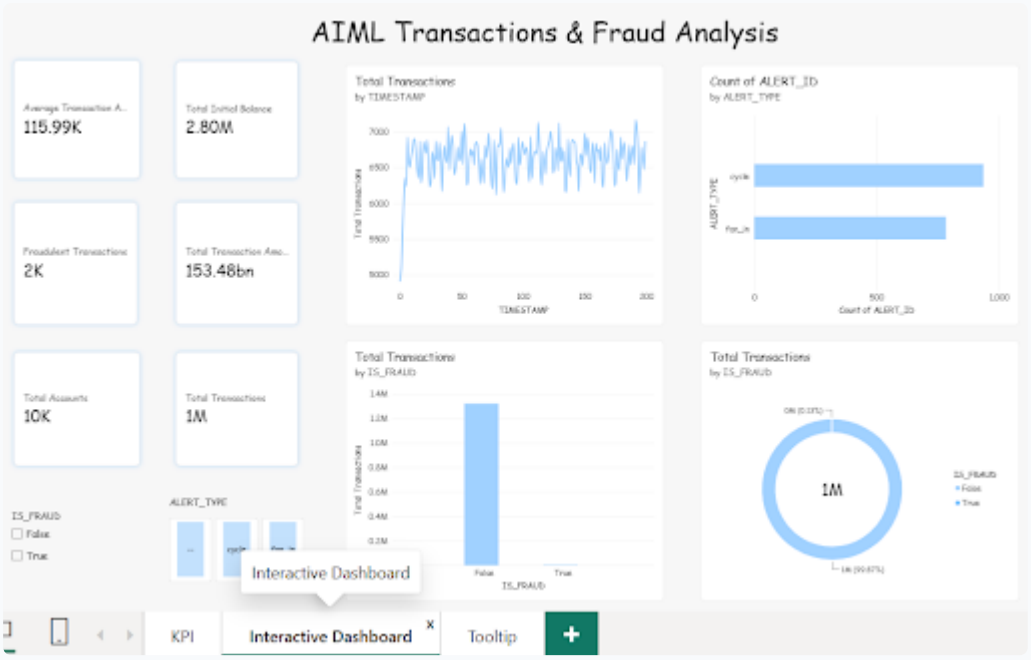


Figure 4. Reference snapshot of primary dashboard layout used for operational audit.

Technical & Data Notes

• All figures are extracted directly from the provided dashboard screenshots; specific non-labeled values represent visual estimations.
• Transaction count formatting truncates total volume to 1M on KPI cards, though detailed calculations confirm ~1,323,23X transfers.
• Current dashboard views do not display customer demographic segments, temporal time units (e.g., minutes vs hours), or machine learning model confidence scores.

by:

Atchaya G